

Graph-Based Methods for Personalized News Recommendation Systems

A Final Project End Semester Report

submitted by

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in partial fulfilment of requirements

for the award of the degree of

BACHELOR OF TECHNOLOGY



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INDIAN INSTITUTE OF INFORMATION TECHNOLOGY,
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May 2026

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I, **Naga Satya Sai Sumakar Kothamasu**, with Roll No: **CS22B1077** hereby declare that the material presented in the Project Report titled **Graph-Based Methods for Personalized News Recommendation Systems** represents original work carried out by me in the **Department of Computer Science and Engineering** at the Indian Institute of Information Technology, Design and Manufacturing, Kancheepuram.

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Naga Satya Sai Sumakar Kothamasu

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This is to certify that the report titled **Graph-Based Methods for Personalized News Recommendation Systems**, submitted by **Naga Satya Sai Sumakar Kothamasu (CS22B1077)**, to the Indian Institute of Information Technology, Design and Manufacturing Kancheepuram, in partial fulfilment of requirements for the award of the degree of **BACHELOR OF TECHNOLOGY** is a bonafide record of the work done by him/her under my supervision. The contents of this report, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

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ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to **Prof. Sivaselvan B** for his unwavering support, invaluable guidance, and thoughtful feedback throughout this project. His deep expertise, insightful suggestions, and constant encouragement have been instrumental in shaping this work, *Graph-Based Methods for Personalized News Recommendation Systems* into a meaningful and impactful contribution.

I extend my heartfelt thanks to the **Department of Computer Science and Engineering, IITDM Kancheepuram**, for providing the academic environment, infrastructure, and technical resources that made this research possible.

I am also deeply grateful to my **professors and mentors** whose teachings, guidance, and inspiration motivated me to explore the fascinating domain of personalized news recommendation systems using Graph Neural Networks. Their continuous support has greatly enriched my learning experience and research journey.

ABSTRACT

The fast development of internet news usage results in the overflow of information and makes it necessary to have a news recommendation system to solve the problem. However, traditional recommendation methods like collaborative filtering and content-based recommendation cannot describe all the relations between user and item properly. Therefore, for improving the recommendation task, we try a graph learning framework with the power of GNNs.

An initial architecture based on the GREP approach was used to obtain interests of both current and possible users. Further development of this model has occurred by improving the approach to representing the data graphically and propagating the interests in such a way. Comparative analysis was conducted between the baseline and improved models.

The experimental results based on the evaluation metrics (MRR, nDCG@5, nDCG@10, and AUC) indicate that the enhanced method significantly outperforms the benchmark model, illustrating the superiority of the graph-based model for personalizing news recommendation. This is consistent with previous studies emphasizing the merits of GNN-based and knowledge-aware recommendation systems.

In summary, this study shows the promising prospects for the application of graph learning for recommendation purposes and lays down groundwork for future improvement, such as integrating more contextual data and ensuring scalability.

KEYWORDS: Graph Neural Networks (GNN); Personalized Recommendation Systems; User Interest Modeling; Ranking Metrics.

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ABBREVIATIONS

ROCAUC	Receiver Operating Characteristic – Area Under Curve
BPR	Bayesian Personalized Ranking
CAUM	Candidate Aware User Modelling
CF	Collaborative Filtering
CNN	Convolution Neural Network
DivHGNN	Heterogeneous Graph Neural Network with Personalized and Adaptive Diversity
GNN	Graph Neural Network
GREP	Graph Neural News Recommendation with Existing and Potential Interest Modeling
KG	Knowledge Graph
LR	Learning Rate
MIND	Microsoft News Dataset
MF	Matrix Factorization
MRR	Mean Reciprocal Rank
NCF	Neural Collaborative Filtering
NRMS	Neural News Recommendation with Multi-Head Self-Attention
nDCG@5	Normalized Discounted Cumulative Gain at rank 5
nDCG@10	Normalized Discounted Cumulative Gain at rank 10
RNN	Recurrent Neural Network
SVD	Singular Value Decomposition

CHAPTER 1

INTRODUCTION

1.1 Problem Background and Motivation

The rapid evolution of digital media technologies has revolutionized the consumption of news as well. Numerous news websites produce an extensive array of articles each day on a wide range of subjects including politics, sports, business, technology, and entertainment, among others. The increased ease with which this process takes place has made it more challenging for consumers to sift through the excess information available.

The idea behind personalized news recommendation systems is that they are designed to resolve this issue through models and delivery of tailored news items based on user preferences. The difference between news recommendation and other kinds of recommendation is that the latter includes several aspects that do not exist when it comes to news recommendation. For example, in news recommendation systems, users' interest changes dynamically, and the life span of the news is limited.

Traditional collaborative filtering systems that depend mainly on interactions between users and items cannot accurately represent the semantics behind the news content and adjust to changing user interests. Even though contemporary deep learning models like attention-based and neural networks have enhanced the representation of content, most of these models ignore the interdependencies between users and news articles.

Graph Learning offers a highly prospective approach through which users and news articles can be considered as nodes on an interaction graph. In this regard, GNNs can play a significant role as they allow representation learning for modeling high-order interactions and connections. The current project focuses on using graph learning approaches to achieve better representation of user news interaction, thereby enhancing personalized news recommendation system performance.

1.2 Scope of the Project

This work aims at designing a news recommender system that relies on graph representation and evaluates its performance using well established ranking metrics. In particular, the model represents user news interactions in a graph format to facilitate the application of GNNs to learn effective representations of users by aggregating information from their neighborhoods.

In the initial stages of the project, a recommendation model will be designed and implemented to form a baseline. Following that, the recommendation model is further improved by making architectural changes that would enable the new model to better cater to the needs and behaviors of the users.

The major emphasis of this project lies more in researching and experimenting with the concepts rather than implementing the solution in practice. The objective is to explore the capabilities of using graph representations for modeling collaboration behavior in dynamic news scenarios and understand the balance between accuracy and efficiency.

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

It is imperative that recommendation systems be seen as essential infrastructure for the Web and be viewed as technologies used to mitigate the issue of information overload through relevant recommendations. Recommendation technology has evolved through various stages from basic heuristic and statistical models to representation learning models and, recently, graph neural network models, where the explicit relationship between users and items is taken into account. News recommendation is unique in its own challenges such as fast content turnover, cold start problems, and user interests, which have necessitated an independent path of research in this area. This chapter provides a survey of existing methods that form the basis of this paper, ranging from classic collaborative filtering algorithms (Sections 2.2), deep learning methods (Sections 2.3), graph neural network-based collaborative filtering (Sections 2.4), and culminating in the three core models used in this paper: LightGCN, GREP, and DivHGNN (Sections 2.5 2.6.2).

2.2 Traditional Collaborative Filtering

In other words, the initial versions of the collaborative filtering (CF) algorithms focused on using past information about users' interactions with items for forecasting their future preferences. In the case of user-based collaborative filtering (UB-CF), users with analogous behavior are clustered, and items that they liked are recommended to other users with the same behavior; similarly, in the case of item-based collaborative filtering (IB-CF), items similar to those previously interacted with are recommended.

Matrix Factorization (MF) enhanced the performance of previous approaches by decomposing the matrix that represents the interaction between users and items into

low dimensional space representations of both. The hidden preferences were captured by these vectors, and the recommendations were made by comparing them. Other approaches such as SVD++ have been used to enhance this approach even more by using the interactions between users. However, MF ignores temporal and correlated behaviors of users and items in its model.

According to Wu et al. [4], the main problem with the MF method is that this method fails to capture the inherent collaborative nature of interactions, i.e., two users that exhibit similar interactions must be mapped into similar vector representations, whereas the loss function for traditional MF does not enforce any constraint in this regard.

2.3 Deep Learning Based Recommendation

2.3.1 Neural Collaborative Filtering

Deep learning introduced non-linearity at the heart of recommendation systems. NCF models have replaced the inner-product used by MFs with multilayer perceptron (MLP) models that model complex, non-linear interactions between users and items, proving that learnt interaction functions performed better than linear models in terms of feedback [4]. The convolutional and recurrent neural networks were then utilized for incorporating textual information of the items and sequential clicks, respectively.

Yet most neural CF models share a fundamental limitation with their MF predecessors: embeddings are derived only from the ID or attribute of the entity itself, without encoding the structure of interaction graph. Hence, two users who never co interact with the same item but are connected through chains of shared interactions remain unrelated in the learned embedding space.

2.3.2 Attention Based News Recommendation: NRMS

News recommendation demands a higher degree of content sensitivity than generic item recommendation, given articles have rich textual content, short lifecycles, and rapidly evolving relevance. Wu et al. [1] suggested the NRMS framework, which applies Trans-

former style self attention at two levels.

Word level: At the word level, MHSA learns the relationship between the words present in the news headline, and this leads to the creation of context-dependent word sequences. These word sequences are further represented as news through additive attention where a relevance score is computed for every word. User level: The same method is applied to the user level where multiple articles are clubbed together in order to create the user vector based on his past articles viewed.

NRMS demonstrated consistent improvements over prior CNN and GRU based news encoders in AUC, MRR, nDCG@5, nDCG@10 on real world news clicks. Its simplicity and strong performance established it as a key baseline for subsequent graph enhanced news recommendation models.

2.3.3 Candidate Aware User Modelling: CAUM

Another limitation of methods like NRMS is user interest is modelled independently of the candidate news being evaluated. Qi et al. [9] says that users typically possess multiple simultaneous interests, and accurate recommendation requires matching the *specific* interest most relevant to a given candidate. Their CAUM approach injects the candidate news representation into the user encoder as a conditioning signal.

CAUM proposes three candidate aware components: (1) a candidate aware self attention network that uses the candidate article as a query to identify globally relevant clicked news; (2) a candidate aware CNN that models local behavioural context by incorporating candidate features into window level convolution; and (3) a candidate aware attention aggregator that weights previously clicked articles by their relevance to the candidate. This single tower architecture simultaneously models user and candidate during inference, achieving improved matching performance over two tower approaches. CAUM highlights the importance of query dependent interest aggregation in news recommendation.

2.3.4 Knowledge Graph Propagation: RippleNet

Knowledge from the external world that is structured could serve as a means to augment the information present in clicks alone. Wang et al. [6] have introduced the idea of **RippleNet**. This method sees the KG as a medium for propagating preference signals. Specifically, for any user, the set of seed preferences (or clicked items) can be augmented by traversing the KG repeatedly: in hop h , the ripple set for a user will be made up of all the KG entities that can be reached using h hops from clicked items. Attention scores of these entities in the ripple set regarding the target item would be calculated using a soft attention mechanism.

2.4 Graph Neural Networks for Collaborative Filtering

2.4.1 Foundations

GNNs extend deep learning to graph structured data through a message passing paradigm: each node iteratively aggregates representations from its local neighbourhood and updates its own state accordingly. After K layers, a node's embedding encodes information from its K hop neighbourhood. Wu et al. [4] and Gao et al. [5] provide comprehensive taxonomies of GNN based recommenders, identifying four core design choices: graph construction, neighbour aggregation, information update, and final representation pooling.

In the recommendation context, users and items are modelled as nodes, and a user item interaction is encoded as an edge. High order connectivity emerges naturally from GNN propagation: a user's embedding after two layers encodes not only their directly interacted items, but also items interacted with by users who share their interaction patterns precisely the collaborative filtering signal that MF was unable to capture properly.

2.4.2 NGCF

Wang et al. [8] proposed NGCF to directly include collaborative signals in the embedding learning process. NGCF builds a user-item bipartite graph and spreads (propagates) embeddings on this graph. The update rule at layer l is:

$$\mathbf{e}_u^{(l)} = \sigma \left(\mathbf{W}_1 \mathbf{e}_u^{(l-1)} + \sum_{i \in \mathcal{N}_u} \frac{1}{\sqrt{|\mathcal{N}_u| |\mathcal{N}_i|}} \left(\mathbf{W}_1 \mathbf{e}_i^{(l-1)} + \mathbf{W}_2 (\mathbf{e}_i^{(l-1)} \odot \mathbf{e}_u^{(l-1)}) \right) \right) \quad (2.1)$$

where $\mathcal{N}_u$ denotes the set of items interacted with by user u , $\mathbf{W}_1$ and $\mathbf{W}_2$ are trainable weight matrices, $\odot$ denotes element wise product, and σ is a non linear activation. The interaction term $\mathbf{e}_i^{(l-1)} \odot \mathbf{e}_u^{(l-1)}$ explicitly encodes first order feature interactions between users and items at each propagation step. Final embeddings are obtained by concatenating representations from all layers. NGCF demonstrated significant improvements over MF and attention based baselines on multiple public benchmarks.

2.5 LightGCN: Simplifying Graph Convolution for Recommendation

NGCF established the value of graph based embedding propagation, He et al. [7] says it has a complex architecture. Through systematic ablation, they found that the two operations adapted from standard GCNs feature transformation matrices and non linear activation functions contribute negligibly to recommendation performance and actively slow down training convergence. Removing both components and retaining only neighbourhood aggregation, they proposed **LightGCN**.

Light Graph Convolution

The propagation rule of LightGCN discards weight matrices and non linearity entirely:

$$\mathbf{e}_u^{(k)} = \sum_{i \in \mathcal{N}_u} \frac{1}{\sqrt{|\mathcal{N}_u||\mathcal{N}_i|}} \mathbf{e}_i^{(k-1)}, \quad \mathbf{e}_i^{(k)} = \sum_{u \in \mathcal{N}_i} \frac{1}{\sqrt{|\mathcal{N}_i||\mathcal{N}_u|}} \mathbf{e}_u^{(k-1)} \quad (2.2)$$

The symmetric normalisation factor $1/\sqrt{|\mathcal{N}_u||\mathcal{N}_i|}$ prevents high degree nodes from dominating the aggregation, similar in spirit to the graph Laplacian normalisation. Only the ID embeddings at layer 0, i.e., $\mathbf{e}_u^{(0)}$ and $\mathbf{e}_i^{(0)}$, are trainable; all subsequent layer embeddings are deterministic functions of these and the graph structure.

Layer Combination

Rather than using only the final layer’s embedding, LightGCN constructs the final representation as a weighted sum over all K layers:

$$\mathbf{e}_u = \sum_{k=0}^K \alpha_k \mathbf{e}_u^{(k)}, \quad \mathbf{e}_i = \sum_{k=0}^K \alpha_k \mathbf{e}_i^{(k)} \quad (2.3)$$

where $\alpha_k = 1/(K + 1)$ by default. This combination serves three purposes: it smoothly integrates neighbourhood signals at different hops, it contains the effect of self connections (avoiding the need for residual connections as an addition), and it removes oversmoothing by retaining a weighted contribution from the original embeddings. The final interaction score is the inner product $\hat{y}_{ui} = \mathbf{e}_u^\top \mathbf{e}_i$, optimised with Bayesian Personalised Ranking (BPR) loss.

LightGCN achieved approximately 16% average relative improvement over NGCF across three public benchmarks under similar experimental conditions. The model’s linear structure also shows analytical insights: He et al. showed that the layer combination is equivalent to a polynomial filter on the normalised graph Laplacian, connecting LightGCN to spectral graph theory.

2.6 Graph Based News Recommendation

The graph based model has been extended specifically to news recommendation, where the interaction graph is augmented with entity, category, and semantic content. Two

models: GREP and DivHGNN are of particular relevance to this work and are detailed below.

2.6.1 GREP: Existing and Potential Interest Modelling

Qiu et al. [2] proposed **GREP** to address a key limitation of content only news encoders: they can't exploit the structural relationships between users and news through the interaction graph, especially for cold start articles that have not yet been widely clicked.

Architecture Overview

GREP constructs a knowledge graph $\mathcal{G}$ from entity co occurrence relationships in news articles and organises the recommendation pipeline into three major components.

The **title encoder** applies MHSA over word embeddings to produce a semantically rich news title representation. Entity embeddings derived from $\mathcal{G}$ via graph attention are then incorporated to produce the full news representation.

The **interest encoder** distinguishes between two types of user interest: *existing interests*, derived from the user's click history, and *potential interests*, inferred by propagating user preferences through the entity knowledge graph to discover related but unobserved content. The interest encoder stacks L alternating layers: each layer updates the existing interest representation by attending over clicked news, and updates the potential interest representation by propagating through the KG neighbourhood of the user's preferred entities. A bi directional interaction layer within each encoder block allows the two interest streams to mutually refine one another.

The **click predictor** computes the final matching score between a user's combined interest representation and a candidate news embedding, trained with a negative sampling cross entropy objective.

Experimental Results

GREP was tested on the large-scale MIND and Adressa datasets. GREP surpassed all the baseline methods like NRMS, KRED, GNUD, and GERL in terms of AUC, MRR, nDCG@5, and nDCG@10. In addition, ablation study results revealed that both interest modules were independent contributors to the performance of GREP. Elimination of the bi-directional interaction layers led to poor results, suggesting that interaction between both interest streams is crucial. Importantly, GREP also surpassed the graph-based models, GNUD, and GERL, where the only interaction considered was that between the user and news items while neglecting the KG, proving the effectiveness of modelling potential interests at the entity level for new articles.

2.6.2 DivHGNN: Heterogeneous Graph Modelling with Diversity

The shortcomings identified by Zhang et al. [3] in previous graph-based news recommendations are twofold: first, the models mainly focus on homogenous graphs, i.e., graphs where nodes and edges are of the same type, and second, they do not consider recommendation diversity, hence generating a list of recommendations that is monotonous. **DivHGNN** solves both problems within one solution framework.

Attributed Heterogeneous Graph

DivHGNN builds an attributed heterogeneous graph with four types of nodes: users, news articles, words, and entities from the knowledge graph, linked by typed edges that denote various relations such as "user news clicked," "news mentioned entities," and "word appeared in news." The graph nodes possess heterogeneous attributes such as textual attributes using GloVe embeddings, knowledge attributes from WikiData embeddings, and topological attributes through Node2Vec embeddings.

Heterogeneous Node Content Adapter (NCA)

Because the three attribute types exist in different embedding spaces and encode semantics at different levels of abstraction, directly aggregating them would introduce

semantic gaps. The NCA addresses this by first projecting each attribute type through a type specific MLP aligner:

$$\tilde{a}_t^v = \text{MLP}_t(a_t^v) \quad (2.4)$$

Cross attribute dependencies are then modelled via self attention over the aligned attribute matrix $\tilde{C}^v$, and the result is fused by a shared MLP into a unified primitive node representation c_v . This design allows the subsequent GNN layers to operate on homogeneous dimensional inputs despite the underlying heterogeneity.

Heterogeneous Graph Representation Learning

After attribute alignment, DivHGNN performs typed neighbourhood aggregation. For each relation type, a relation specific attention network assigns importance weights to neighbour nodes. The aggregated neighbourhood signal is combined with the node's own representation through a gating mechanism. Multiple relation specific aggregations are merged via a type level attention layer that learns the relative importance of each edge type for the target node.

Variational Representation and Diversity Modelling

To model the inherent uncertainty and diversity of user interests, DivHGNN employs variational representation learning: node embeddings are parameterised as Gaussian distributions $\mathcal{N}(\mu_v, \sigma_v^2)$ rather than point vectors. Sampling from these distributions during training regularises the representation space and encourages the model to encode a broader coverage of interests, which is then exploited at inference time through Personalised and Adaptive Diversity (PAD) modelling. PAD balances relevance and diversity in the final ranked list by incorporating individual level diversity objectives that penalise redundancy within a user's recommendation list whilst still rewarding accuracy.

Experimental Results

DivHGNN was evaluated on the MIND dataset and demonstrated improvements over GREP, NRMS, and other graph based baselines in both AUC, nDCG and diversity metrics, showing that the heterogeneous graph architecture and variational diversity modelling complement each other.

2.7 Comparative Analysis and Research Gaps

Table 2.1: Comparative summary of key recommendation models reviewed

Model	Graph Based	Heterogeneous	Content Aware	KG Enriched	Diversity
MF / SVD++	No	No	No	No	No
NRMS	No	No	Yes	No	No
CAUM	No	No	Yes	No	No
RippleNet	Yes	Yes	No	Yes	No
NGCF	Yes	No	No	No	No
LightGCN	Yes	No	No	No	No
GREP	Yes	Limited	Yes	Yes	No
DivHGNN	Yes	Yes	Yes	Yes	Yes

The survey highlights the evolution of recommendation models from matrix factorization and content-aware deep learning approaches to graph-based and heterogeneous graph models. While LightGCN and NGCF are effective general recommendation models, they do not utilize the rich textual information available in news articles. GREP addresses this by combining content and graph structures, but it does not consider recommendation diversity. Similarly, DivHGNN improves diversity modeling but introduces high computational complexity, making large-scale deployment challenging.

This work is motivated by GREP’s integration of content embeddings with graph-based interest propagation and LightGCN’s simplified propagation strategy. The objective is to design and evaluate a graph neural recommendation model that balances expressive power and computational efficiency within a homogeneous graph framework for comparison against existing news recommendation baselines.

CHAPTER 3

METHODOLOGY

3.1 System Overview

The proposed system architecture involves a modular pipeline involving deep learning based on the GREP algorithm, making it feasible for the pipeline modules to interact with one another efficiently while scaling and being flexible enough. Firstly, the pre-processing stage entails pre-processing of data through creation of vocabulary, entity-word embeddings, and construction of entity co-occurrence graph in news articles.

As the model runs, it receives input batches consisting of users' internet history and also the different news items that form the candidates. Each one of the news articles is represented using vectors, which the algorithm uses in order to assign scores for relevance, which reflects the possibility of the user clicking on the news article. This system is designed to capture the dynamism of users' reading interest.

The central components of model include Title Encoder and stack of Interest Encoding Blocks. The Title Encoder makes use of multi-head self-attention along with gated aggregation techniques to derive meaningful context from the titles of news articles. Subsequently, these semantic embeddings are passed through several layers of interest encoding blocks to learn user behavior from various angles.

Specifically, the interest encoding module performs:

- existing interest modeling through attention mechanisms applied over the user's browsing history,
- potential interest modeling using a graph transformer operating on entity embeddings to uncover latent preferences, and
- a uni-directional interaction mechanism that effectively combines textual representations with graph-based semantic information.

On the other hand, the model separately encodes both the users and the candidate news articles, making use of the features at the title level and entity level using adaptive gate functions. Lastly, the relevance score is calculated by means of a dot-product calculation involving the encoded user and news article embeddings, allowing for effective ranking of the candidate news articles.

In contrast with the classical GREP paradigm, some significant advancements have been made in terms of the developed system. This involves the incorporation of time-decay weights to prioritize more recent activities of the users, the utilization of sparse graph algorithms to improve efficiency in computations, and advanced fusion methods using attention mechanisms. As a whole, all of these advancements contribute to the development of an efficient Transformer-GNN architecture.

3.2 Dataset Description

MIND Dataset, which is one of the largest benchmark datasets for news recommendation studies, has been utilized for training our model. MIND Dataset has a lot of data points that are helpful in developing content-based recommendations and behavior-based recommendations as well.

3.2.1 MIND Small

MIND-Small is part of the MIND dataset that is used for fast prototyping and experimentation purposes. It offers an accessible dataset size without losing the properties of real-world news recommendations datasets.

3.2.2 MIND Large

MIND-Large is the complete dataset that is utilized to train and evaluate the model. The size of the dataset is increased considerably, thus making it appropriate for assessing the scalability of the model.

Table 3.1: Overview of MIND-Small Dataset

Metric	MIND-Small
# Users	50,000
# News	51,282
# Entities	26,904
# Avg. Entities	3.03
# Categories	17
# Subcategories	264
# Avg. History Length	32.54
# Avg. Title Words	10.75
# Positive Samples	236,344
# Negative Samples	5,607,100

Table 3.2: Overview of MIND-Large Dataset

Metric	MIND-Large
# Users	7,11,222
# News	1,01,527
# Entities	42,007
# Avg. Entities	3.16
# Categories	18
# Subcategories	285
# Avg. History Length	32.9
# Avg. Title Words	10.66
# Positive Samples	33,83,656
# Negative Samples	8,01,23,718

3.3 Implementation Details

The suggested model employs the framework of PyTorch and incorporates some components which would be working on the task of learning the representation of the user in a holistic manner considering both his explicit and implicit likes/dislikes.

3.3.1 GREP Small

The lightweight GREP variant reduces computational complexity by using fewer attention heads, simpler graph propagation, and fewer stacked interest layers while maintaining an embedding dimension of 100. This design preserves the core structural modeling capability of GREP while achieving faster training and lower memory consumption.

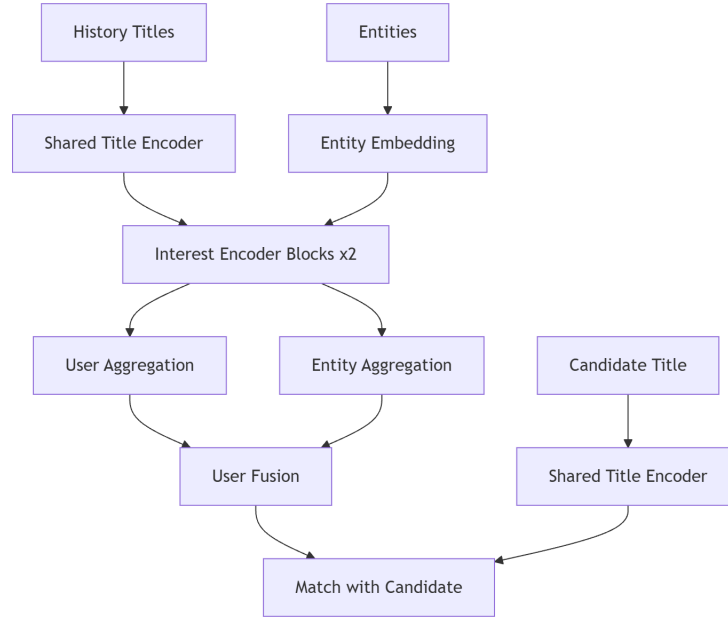


Figure 3.1: GREP Small

3.3.2 GREP Large

This implementation includes multi-head self-attention for title encoding, two stacked Interest Encoding Blocks, a Graph Transformer Layer for potential interest modeling, bi-directional title–entity interaction, and gated fusion for user and news representations. Training uses cross-entropy loss to approximate the InfoNCE objective, where the positive candidate is placed at index 0 among $k + 1$ candidates. Additional stabilization techniques include gradient clipping (max norm = 1.0), NaN gradient replacement with zero, and reduced learning rate scaling.

3.3.3 GREP Improved

GREP architecture is a hierarchical modular architecture that contains the encoding of the news content and the user interface within the system. This architecture consists of three major modules, which are as follows:

Title Encoder

News Headline Encoder encodes the headlines of the news to semantic vectors by using self-attentions. News Headline Encoder employs multi-head self-attentions to convert

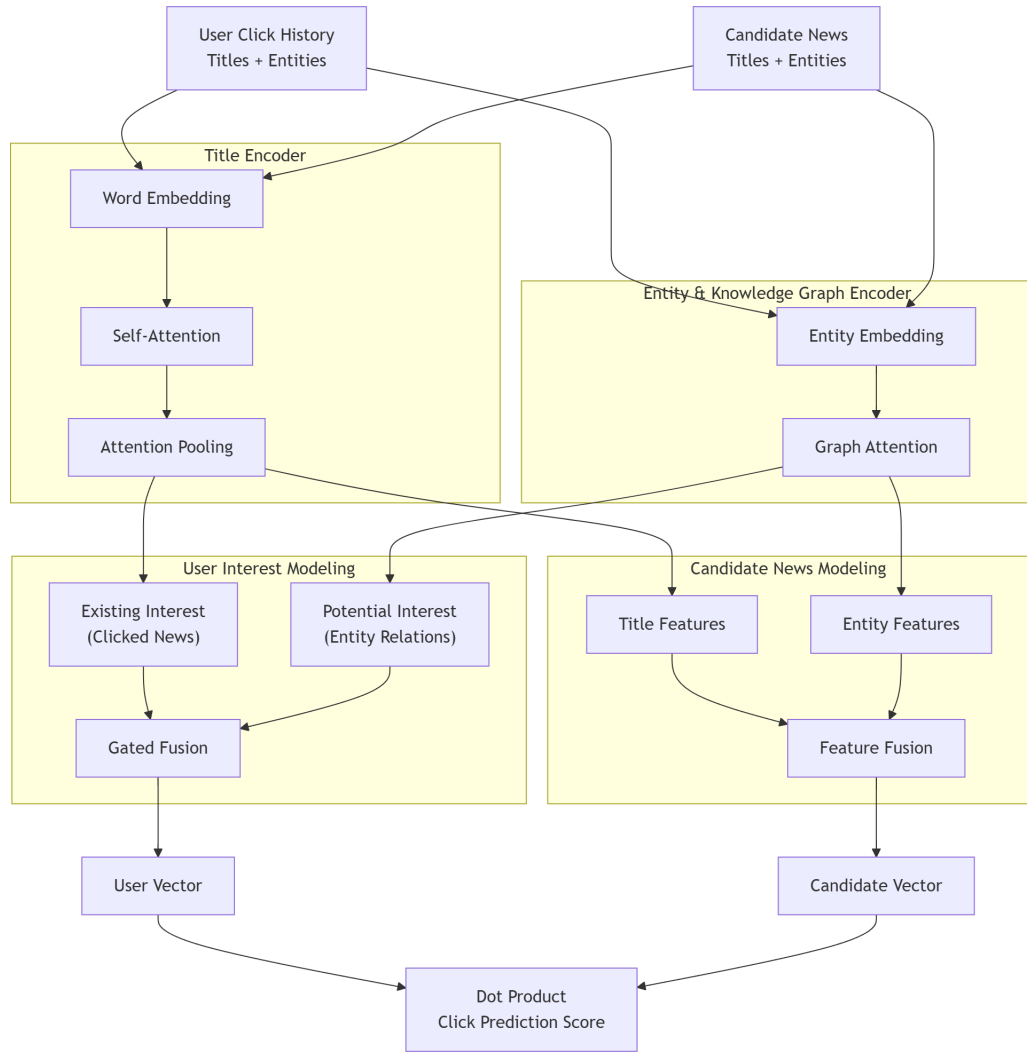


Figure 3.2: GREP Improved

every news headline into semantic vectors according to the relationship between the words in the headlines. Residual connections and layer normalization play an important role in ensuring stability during training while preserving information from every word. Gated Aggregation offers weight values for the words in the headlines, thus converting every news into fixed vectors.

Interest Encoder

The Interest Encoder is the core component responsible for modeling user preferences through two complementary pathways:

- **Modeling of Existing Interests:** Uses multiple self-attention heads over the representation of the user's history of clicked news titles in order to model inter-

actions among different clicked news titles and similar topics, resulting in more precise representations of the titles with internews interaction modeling.

- **Modeling of Potential Interests:** Uses GNN to understand the knowledge graph of entity occurrence. GNN modeling for an entity uses the information from other entities connected to it through attention mechanism, allowing multi-hop interactions. This solves the cold start problem, allowing users' hidden interests to be revealed.
- **Bi-directional Interaction Model:** Works together to integrate and match the representation of the user's existing and potential interests through the use of a bipartite graph that links both news and entities. This model ensures knowledge propagation in two directions, since the potential interests provide knowledge at the level of the entities in the titles.

The multi-block nature of the design allows for successive enhancement of the representation of users via successive interactions on the graph, thereby allowing deep reasoning on the graph structure and successive enhancement of learned features.

Prediction and Aggregation

Both of these can be represented using a dynamically generated vector from a combination of the news title and the entity information in the title using the process of gating. In order to compute the final score of the similarity between them, the dot product similarity score between them is computed, which results in a single number.

The system makes many advancements on the baseline GREP framework through its incorporation of temporal decay weight for modeling the recent-user and graph sparsification method for computational efficiency, thus yielding an optimized Transformer-GNN model for the task of news recommendation.

3.4 Architectural and Algorithmic Improvements

Apart from the base paper on GREP, there are a number of improvements made in the model that can assist in improving it from an efficiency point of view.

3.4.1 Sparse Knowledge Graph Computation

Motivation: The original GREP paper applies the graph transformer uniformly over the entire entity set, resulting in computational complexity $O(|E|)$, where $|E|$ is the total no. of entities in the knowledge graph. This becomes prohibitive for large knowledge graphs.

Proposed Approach: The improved implementation processes only batch-relevant entities and their neighbors, building a dynamic subgraph per batch. This reduces complexity from:

$$O(|E|) \rightarrow O(|E_{\text{batch}}|)$$

where $|E_{\text{batch}}|$ represents the set of entities present in the current batch plus their direct KG neighbors. This approach maintains model expressiveness while significantly reducing memory requirements and computation time, making the model scalable to large knowledge graphs.

3.4.2 Temporal User Modeling

Motivation: User interests change with time. Current behavior can be expected to capture current interests better than past behavior. The initial baseline GREP does not account for the passage of time within the interaction history.

Proposed Approach: A recency weighting mechanism is applied to user interactions:

$$w_i = \exp(-\lambda \cdot (\text{MAX_HIST} - i))$$

where w_i is the weight for the i -th historical interaction, λ is the temporal decay factor, and MAX_HIST is the history length.

Implications: This model encompasses short- and long-term changes in interest. Recently occurring events have more weight, meaning that the effect of past preferences diminishes. This is very important for real-life use cases, where users' interests may drastically change within days or even hours. The value of $\lambda = 0.005$ was selected to create mild weights without neglecting old events.

3.4.3 Explicit Hard Negative Sampling

Motivation: Random negative sampling, the baseline approach, draws negatives uniformly from all non-clicked items. However, the most informative negatives are “hard negatives,” which are items similar to positive samples but not clicked by the user, as these challenging examples improve ranking discrimination.

Proposed Approach: The training process adopts a two-tier negative sampling strategy combined with curriculum learning, where hard negatives are selected from the same category as the positive sample to provide stronger learning signals, while random negatives are drawn from different categories to ensure training stability with easier examples. The proportion of hard negatives is gradually increased during training according to the following schedule:

$$\text{hard_neg_ratio} = 0.3 + 0.4 \times \frac{\text{epoch} - 1}{\text{NUM_EPOCHS} - 1}$$

This progressively raises the difficulty level from 30% to 50% hard negatives across epochs. A category-indexed lookup table is used to enable efficient $O(1)$ hard negative sampling.

3.4.4 Data Augmentation Pipeline

Rationale: Data augmentation helps in enhancing robustness and generalization by introducing variability in the training data. **Approach:** The suggested pipeline leverages a number of techniques for augmentation, which include **Title Dropout**, in which 30% of title tokens are randomly dropped in order to obtain robust embeddings, **History Cropping**, in which interaction histories with varying window sizes are leveraged for generalization, and **Entity Enrichment**, where 1-hop knowledge graph neighbors are added to enrich semantic information. Augmentation is applied to each training example at a rate of 40%.

3.5 Training Strategy Enhancements

3.5.1 Curriculum Learning

The process of curriculum learning is characterized by the progressive rise of training complexity rather than presenting all degrees of training complexity simultaneously. In this regard, the model has the opportunity to initially learn basic patterns and then adjust to more complex examples.

Curriculum learning in the context of GREP Improved is achieved through the use of parameter freezing and the hard negative ratio. In the first two epochs, word embeddings are frozen so that the structure parameters and the entity embeddings become stable and subsequently all the parameters are optimized together. Also, during this period, the hard negative ratio increases from 0.3 to 0.5.

3.5.2 Mixed Precision Training (Automatic Mixed Precision)

In mixed precision training, both fp16 and fp32 calculations are used to optimize the performance of the training process without compromising stability. In PyTorch AMP, majority of the calculations take place in fp16, and there is an implementation of loss scaling and gradient clipping techniques to ensure the training process remains stable. There is approximately 50% lower memory usage and 2x faster training.

3.5.3 Gradient Accumulation

Gradients are accumulated to allow for the use of larger batch sizes than the GPU would otherwise allow because instead of updating weights on each individual batch, they accumulate gradients until there is enough data to update the weights. With ACCUMULATION_STEPS = 2, two batches of size 256 are processed to achieve an effective batch size of 512. The scaling of the loss prior to backpropagation maintains stability in the updating of gradients.

3.5.4 Gradient Clipping and NaN Handling

The use of gradient clipping ($\|g\|_2 \leq 1.0$) in addition to replacing any NaN gradients by zeros and the replacement of $-\infty$ masking value by large negative values ($-1e^4$) ensures training stability. The above modifications address instability issues in gradient computation, particularly when using mixed precision and attention mechanisms.

3.5.5 Learning Rate Scheduling

Motivation: Learning rate scheduling improves convergence by using larger updates early in training and smaller updates later.

Implementation: Training uses a two-stage strategy. First, linear warmup over 500 steps gradually increases the learning rate:

$$\text{lr}_{\text{warmup}}(t) = \text{lr}_{\text{base}} \times \frac{t}{\text{WARMUP_STEPS}}$$

After warmup, the learning rate decays exponentially each epoch:

$$\text{lr}_{\text{epoch}} = \text{lr}_{\text{base}} \times (\text{LR_DECAY_FACTOR})^{\text{epoch}-1}$$

where $\text{LR_DECAY_FACTOR} = 0.95$.

3.5.6 Early Stopping

Early stopping helps to prevent overfitting by checking the validation AUC after each epoch. When there is an improvement, the best model is retained; if not, the number of patience steps increases. The training process ends when there is no improvement for two successive epochs.

3.6 Loss Function: Cross Entropy

The baseline GREP uses softmax cross-entropy over the $k+1$ candidates (1 positive + k negatives):

$$\mathcal{L}_{\text{CE}} = -\log \frac{\exp(s_0)}{\sum_{j=0}^k \exp(s_j)}$$

where s_0 is the positive candidate score and s_j are the negative candidate scores. This approximates the InfoNCE objective and provides stable, well-distributed gradients.

3.7 Optimization and Efficiency Improvements

3.7.1 Batch-wise KG Filtering

Only nodes associated with the present batch and their immediate neighborhoods are subjected to transformation using the Graph Transformer, thus avoiding unnecessary computations without compromising the expressive power of the model.

3.7.2 Memory-Efficient Tensor Construction

Several memory optimizations are applied, including pre-computing neighbor masks, using sparse representations where possible, minimizing tensor copies through in-place operations, and registering KG neighbor tables as buffers instead of parameters. Together, these techniques reduce peak memory usage by approximately 30–40% without sacrificing model capacity.

3.7.3 Reduced Learning Rate for Entity Embeddings

Motivation: Entity embeddings are initialized from pre-trained knowledge graph embeddings, which contain useful semantic information that should be preserved during recommendation training.

Implementation: Entity embeddings use a 10x smaller learning rate (0.00001) than

other model parameters (0.0001). This enables task-specific fine-tuning while maintaining the semantic structure of the original embeddings, improving generalization and training stability.

3.8 Experimental Setup

Table 3.3: Key Hyperparameters Used in the experiment

Hyperparameter	Value	Reason
HIDDEN_DIM	100	Latent dimension for all modules.
TITLE_HEAD_NUM	5	Enables MHA
INTEREST_LAYER_NUM	2	Allows deeper user interest modeling.
KG_HEAD_NUM	4	Models graph relations
MAX_HISTORY_LEN	50	Sufficient user history coverage.
NEG_SAMPLE_RATIO	6	Balances positive vs negative samples.
BATCH_SIZE	256	Training with efficient GPU usage.
LEARNING_RATE	0.0001	Ensures stable convergence.
NUM_EPOCHS	10	Allows sufficient training.
DROPOUT	0.3	Regularization to prevent overfitting.
WEIGHT_DECAY	0.0001	L2 regularization.
TEMPORAL_DECAY_LAMBDA	0.005	Weights recent interactions higher.
MAX_ENTITY_NEIGHBORS	20	Controls KG complexity.
Device	Cuda	Faster computation

Evaluation metrics used: AUC, MRR, nDCG@5, nDCG@10

CHAPTER 4

RESULTS AND ANALYSIS

4.1 Results

Different models were trained on MIND-Small and MIND-Large datasets and the performance metrics obtained are shown in the below table.

Table 4.1: Comparison of different models

Model	Trained on	AUC	MRR	nDCG@5	nDCG@10
GREP Small	MIND-Small	59.16	30.17	28.47	34.93
LightNRMS	MIND-Small	60.17	31.09	29.13	35.68
GREP Large	MIND-Small	62.77	33.74	31.82	38.25
LightNRMS(Glove)	MIND-Small	64.72	35.21	33.21	39.85
GREP Improved	MIND-Small	66.65	37.14	35.63	41.73
GREP Improved	MIND-Large	68.05	38.07	36.56	42.83
GREP (Paper)	MIND-Large	69.44	34.40	37.54	43.22

4.2 Observations

Results obtained from the experiment show clearly that the GREP Improved model always outperforms the benchmark models based on different metrics used for comparison. For example, on the MIND-Small dataset, the model attained an AUC value of 66.65% while the benchmark GREP Small model achieved only 59.16%. The performance of the proposed model is also better than the LightNRMS model with GloVe embeddings whose performance was 64.72%.

The reason for this progress is the incorporation of the temporal decay feature. With large weight being assigned to those news articles that have been viewed lately by a particular user via $\text{weight}[i] = \exp(-\lambda \cdot (MAX_HIST - i))$, the model successfully accounts for the recency aspect of user activity. It perfectly correlates with reality, as user recent activity usually demonstrates his or her recent interests better than old one.

The utilization of MHSA in the title encoder and graph transformer leads to better capturing of various semantics and user preferences, thus providing another increase in performance of about 4.93% compared to GREP Large.

The bi-directional relationship between entities and titles leads to enhanced ranking performance, as shown by the nDCG@5 and nDCG@10 metrics with values of 35.63% and 41.73%, respectively. Besides, the model has scalability for larger data sets, as seen from the AUC score of 68.05% on the MIND-Large data set.

4.3 Analysis

GREP Improved’s better performance is mainly due to a few architectural advancements. First, the Graph Transformer uses multi-head attention between neighboring entities rather than simply aggregating entities for more complex entity representations. Second, the FuseGate module enhances representation learning using adaptive gating of titles and entities.

However, due to efficiency purposes, calculations are performed on the required entities for the current batch along with their neighbors and no calculations are wasted on other entities. Additionally, for more stable training, it is suggested that instead of using $-\infty$, very high negative values be used for masking, to avoid any NaN gradients in the mixed precision training case.

The full suite of optimization techniques described in Chapter 3, such as sparse knowledge graph computations, temporal user modeling, hard negative sampling, and data augmentation, all come together to drive performance gains. Training method optimizations like curriculum learning, mixed precision training, gradient accumulation, learning rate scheduling, and early stopping add even more stability and convergence improvements. This synergy of optimizations leads to a 4.93% gain in AUC from GREP Large (62.77%) to GREP Improved (66.65%) on MIND-Small and a 5.28% gain on MIND-Large (68.05% vs. 62.77%).

4.4 Current Limitations

Nonetheless, even though the performance boost is remarkable, it still cannot be denied that there are some drawbacks in the GREP Improved framework, making it not feasible for real-world application. Despite the improved version of the framework achieving the AUC score of 68.05%, it fails to match the 69.44% obtained from the experiment conducted by the authors. This could imply the necessity to optimize the model for better performance.

- **High Computational Cost:** The use of multi-head attention mechanism, graph transformers, and stack of interest encoders greatly increases the computational cost of the model.
- **High Memory Cost:** A large amount of GPU memory is required to train the model due to embedding storage and neighbor extraction from the knowledge graph.
- **Tuning of Parameters:** There is strong dependence on the selection of hyper-parameters such as attention heads, dropout rate, time decay factor, and interest encoders' layers.
- **Dependence on Knowledge Graph Quality:** Incomplete or inaccurate knowledge graphs can reduce the effectiveness of entity-aware recommendation learning.
- **Fixed User History Length:** Using a fixed interaction history may not accurately capture different user behavior patterns, particularly between short-term and long-term interests.
- **Limited Interpretability:** Complex components such as multi-head attention and gated fusion improve accuracy but make it difficult to explain which entities or interactions influence recommendations.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

The development of a better news recommendation system using a GNN is accomplished with this project by overcoming the main problem in modeling users' preferences in news recommendation systems. This project begins from the basis of collaborative filtering and gradually moves on to more advanced research on graph neural networks, leading to the proposed enhancement approach.

5.2 Key Contributions

The paper proposes an improved version of the GREP architecture with several design and optimization enhancements. One major improvement is the incorporation of a temporal decay factor in the weighting function which helps the model emphasize recent user interactions and better capture changing user interests in news recommendation systems.

Additional improvements focus on scalability and stability. Sparse entity computation limits graph operations to entities relevant to the current batch and their KG neighbors, reducing complexity to $O(|batch_ent| \cdot K)$ while preserving representational power. Numerical stability is also improved by replacing $-\infty$ masking values with $-1e^4$, preventing NaN gradients during fp16 training.

The experimental results obtained from the experiments conducted on the MIND-Small and MIND-Large datasets prove the efficiency of the proposed model. In this regard, the model attains an AUC score of 66.65% for MIND-Small data while surpassing GREP Small model by 7.48% and an AUC score of 68.05% for MIND-Large data. Excellent ranking ability is evidenced by nDCG@5 score of 35.63% and nDCG

5.3 Future Scope

- **Adaptive History Modeling:** Adaptively adjust the length of browsing history according to each user to take into account both their short-term and long-term preferences.
- **Hierarchical Knowledge Graph Modeling:** Build knowledge graphs at multiple levels, including entities, concepts, and categories.
- **Contrastive Learning:** Enhance the representation of each user by bringing similar users closer together while pushing dissimilar users apart.
- **Advanced Fusion:** Experiment with various fusion mechanisms, like *FuseGateAttention*, to more effectively fuse textual and graph data.
- **Scalability and Efficiency:** Utilize techniques like graph compression, graph pruning, graph quantization, and model distillation to lower memory and inference costs.
- **Adaptive Temporal Decay and Multi-Task Learning:** Research adaptive temporal decay models and auxiliary tasks, such as category prediction and entity linking.
- **Interpretability:** Leverage attention visualization and explainable recommendation approaches to increase interpretability and user satisfaction.

5.4 Final Remarks

As mentioned above, the discussed work proves that the combination of graph neural networks and attention and temporal approaches helps to incorporate interest dynamics into news recommendation systems effectively. At the same time, GREP Improved demonstrates good empirical results and provides different opportunities for optimization.

Nevertheless, there are many issues in terms of the studied case, such as high computational complexity, scalability problems, lack of transparency, and necessity to create high-quality knowledge graphs. Despite all these disadvantages, the suggested approach may be considered a great starting point for future research in the sphere of personalization systems.

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Weekly Review Report

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3	19/01/2026 - 25/01/2026	Datasets Exploration and Understanding <i>Sush</i> 24/1/26	
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Weekly Review Report

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13	30/03/2026 - 05/04/2026		
14	06/04/2026 - 12/04/2026		
15	13/04/2026 - 19/04/2026		
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